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Advanced postoperative tissue antiadhesive membranes enabled with electrospun nanofibers

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Tissue adhesion is one of the most common postoperative complications, which is frequently accompanied by inflammation, pain, and even dyskinesia, significantly reducing the quality of life of patients. Thus, to prevent the formation of tissue adhesions, various strategies have been explored. Among these methods, placing anti-adhesion membranes over the injured site to separate the wound from surrounding tissues is a simple and prominently favored method. Recently, electrospun nanofibers have been the most frequently investigated antiadhesive membranes due to their tunable porous structure and high porosities. They not only can act as an essential barrier and functional carrier system but also allow for high permeability and nutrient transport, showing great potential for preventing tissue adhesion. Herein, we provide a short review of the most recent applications of electrospun nanofibrous antiadhesive membranes in tendons, the abdominal cavity, dural sac, pericardium, and meninges. Firstly, each section highlights the most representative examples and they are sorted based on the latest progress of related research. Moreover, the design principles, preparation strategies, overall performances, and existing problems are highlighted and evaluated. Finally, the current challenges and several future ways to develop electrospun nanofibrous antiadhesive membranes are proposed. The systematic discussion and proposed directions can shed light on ideas and guide the reasonable design of electrospun nanofibrous membranes, contributing to the development of exceptional tissue anti-adhesive materials in the foreseeable future.

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1. Introduction

Postoperative tissue adhesions are highly prevalent, arising from processes of fibrous tissue exudation, macrophage infiltration, and excessive proliferation of fibroblasts, which are considered to be closely associated with surgical incisions, trauma and other pathological situations.¹ Clinically, tissue adhesions typically encompass various anatomical sites, including tendon, dural sacs, the peritoneum, and pericardium.^{2–5} Apart from giving rise to prevalent symptoms such as pain, impaired movement, and paralysis, postoperative tissue adhesions may also have profound consequences on specific organs. For instance, undesirable postoperative adhesions can result in complications such as intestinal obstruction, female infertility, heart failure, and even threaten the

patient's life.^{6–9} Furthermore, the issue of adhesions also causes a significant economic burden and consumption of healthcare resources worldwide.^{10,11}

Currently, various approaches have been employed to manage tissue adhesions in clinical practice, among which surgical release is the most common. However, the efficacy of this method is limited, resulting in patient discomfort and imposing economic pressure. Therefore, attention has gradually shifted towards anti-adhesion membranes with physical barrier properties.¹² Currently, some anti-adhesion products used in clinical practice have been approved by the US Food and Drug Administration, which mainly include Interceed® (oxidized regenerated cellulose), Seprafilm® (hyaluronic acid-carboxymethyl cellulose), and Adept® (4% icodextrin solution).^{13–15} These products can exert a certain anti-adhesion effect in the corresponding tissue adhesions, such as tendons and peritoneum. However, despite their clinical use, there are still many problems. For example, Interceed® cannot completely prevent adhesions after blood infiltration, and thus all the blood in the surgical area must be cleared before use, increasing the difficulty of surgery.¹⁶ Seprafilm® degrades quickly, limiting its ability in reducing adhesions.¹⁷ Additionally, Adept® may cause allergic reactions in patients.¹⁵ Therefore,

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there is an urgent need for new types of more optimal anti-adhesion materials.

Recently, advances in materials science have expanded the choice of materials for creating these membranes, such as hydrogels, polymer films, and human amnion.^{18–25} However, all these materials exhibit certain drawbacks. For example, hydrogels are easy to break and degrade promptly *in vivo* due to their inadequate mechanical properties.^{26,27} The structure of polymer films is usually dense and non-porous, impeding the exchange of nutrients and waste.¹³ Additionally, the application of human amniotic membranes may potentially induce immune rejection responses.²⁸ Also, considering the performance, cost and complex fabrication processes for these materials, further advances in tissue anti-adhesion using anti-adhesive membranes are limited.

Electrospinning, as a versatile, low-cost and large-scale method to produce biomedical nanofibrous membranes with extraordinary properties, has been widely used in the biomedical field, such as tissue-engineered scaffold, wound dressings, drug carriers, biosensor, sound absorption, and filtration.^{29–40} During electrospinning, high-voltage electrostatic forces act on solutions or melts to generate nanofibers, offering several prominent features of large surface to-volume ratio, adjustable porosity and tunable nanofiber composition for desired properties and functions.^{41,42} The nano-micro porous structures of nanofibrous membranes enable nutrient diffusion, while preventing cell penetration, thereby averting postoperative adhesions. Moreover, nanofibrous membranes also exhibit favorable mechanical properties and flexibilities, which can facilitate their attachment to the surgical site.⁴³ More importantly, nanofibrous membranes are expected to emulate the biological environment and influence cellular behavior due to their natural mesh-like 3D architectures that mimic the natural extracellular matrix (ECM).^{44,45} Benefiting from these integrated properties, nanofibrous membranes not only can reduce tissue adhesions effectively but also foster tissue repair, making them promising prospects for anti-adhesion applications.

Recent years have witnessed a growing interest in developing advanced anti-adhesion materials by using electrospun nanofibers. Also, substantial advancements have been made through continuous efforts and investigation by researchers. Thus far, more than 200 publications (searched in Web of Science with keywords of electrospinning & adhesion prevention) regarding electrospun nanofibrous antiadhesive materials have been reported. However, to date, there is no systematic and comprehensive review focusing on the introduction of electrospun nanofibrous anti-adhesion membranes.^{46–50} To the best of our knowledge, the existing reviews are predominantly centered on anti-adhesive materials that are specific to a certain human tissue such as tendons.^{51,52} Consequently, an in-depth exploration of adhesion across various parts of the human body and the shared or different characteristics of anti-adhesion membranes remains limited. Due to the important role of electrospun nanofibers in the process of tissue regeneration, as well as

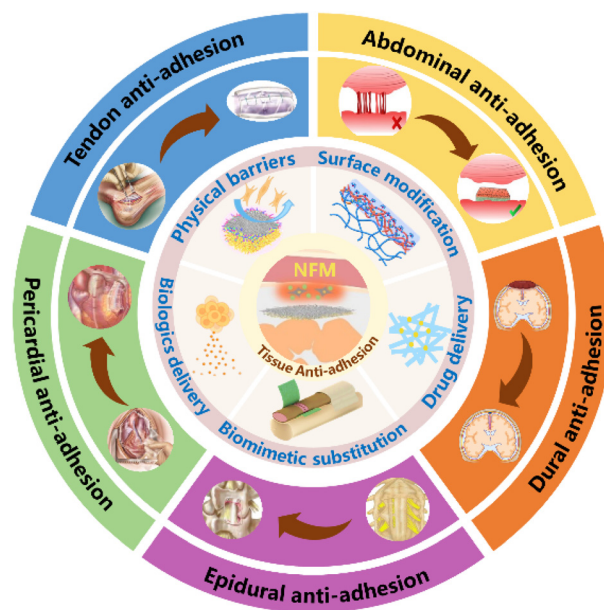


Fig. 1 Introduction to the characteristic tissue anti-adhesion applications of electrospun nanofibrous membranes (NFMs). Reprinted with permission from ref. 51. Copyright 2017, Elsevier. Reprinted with permission from ref. 131. Copyright 2022, Elsevier. Reprinted with permission from ref. 53. Copyright 2022, Elsevier. Reprinted with permission from ref. 135. Copyright 2022, Springer Nature. Reprinted with permission from ref. 47. Copyright 2020, the American Chemical Society.

their tremendous progress in recent years, a timely and comprehensive review on anti-adhesion nanofibrous membranes is imperative.

This review focuses on electrospun nanofibrous membranes for tissue anti-adhesion applications toward human body parts in tendons, peritoneal regions, dural sacs, pericardium, and meninges, which is summarized in Fig. 1. Firstly, the mechanisms and factors leading to tissue adhesion are overviewed in each section. Then, the synthesis strategies and structure/surface/composition design concepts of anti-adhesion nanofibrous membranes from single fibers to nanofiber aggregation are emphasized. We also highlight the performance advantages and existing potential problems of each membrane used in different tissues. Finally, the challenges and future opportunities of antiadhesive nanofibrous membranes are discussed. We hope that this review illustrates different strategies to design, fabricate, and use nanofibers for tissue anti-adhesion applications and inspires researchers to tackle future challenges in this field.

2. Mechanism of postoperative tissue adhesion and the design criteria for anti-adhesive materials

Adhesion formation is a complex process. This section provides an overview of the mechanisms involved in adhesion for-

mation at both the cellular and tissue levels, which is particularly important for the design of subsequent anti-adhesion strategies.

At the cellular level, mesothelial cells, immune cells, and fibroblasts play crucial roles in postoperative tissue adhesion. Following tissue damage, these cells interact and collaborate, promoting the formation of adhesions. Mesothelial cells, which can secrete tissue plasminogen activator (tPA) and plasminogen activator inhibitor (PAI), constitute a protective membrane that covers the surfaces of the peritoneum and pericardium.⁵⁴ The release of tPA promotes the fibrinolysis process, while the action of PAI inhibits protein deposition.⁵⁵ When mesothelial cells are injured, there is an increase in PAI concentration and a decrease in tPA levels, resulting in reduced fibrinolytic activity and the destruction of the balance between coagulation and fibrinolysis. Consequently, the disrupted equilibrium leads to adhesion formation.^{56,57} Immune cells (including neutrophils, macrophages, and mast cells) and their released mediators significantly contribute to the regulation of inflammation, cell migration, proliferation, angiogenesis, and tissue remodeling, influencing the formation of adhesions.^{58,59} In the initial phases of healing, neutrophils and their oxygen free radicals emerge as critical factors contributing to early inflammation.⁶⁰ Neutrophils accumulate significantly in the injured area, contributing to the production of reactive oxygen species (ROS). This highly reactive ROS is closely associated with oxidative stress, subsequently impacting tissue adhesion. Blocking the production of highly reactive oxygen radicals by neutrophils, rather than eliminating these cells, contributes to preventing adhesion formation resulting from early postoperative inflammatory reactions.⁶¹ Research has confirmed that employing anti-neutrophil serum (ANS) and clearing methods for ROS can effectively reduce adhesion occurrence.⁶² Fibroblasts are the main cellular components of loose connective tissue, playing a pivotal role in tissue repair. During wound healing, fibroblasts proliferate through mitosis, synthesizing collagen fibers and matrix components, which form granulation tissue to fill the defect.⁶³ Tendon and epidural adhesions are thought to be closely related to the excessive proliferation of fibroblasts. Tendon adhesions often result from an imbalance between endogenous and exogenous healing. Exogenous healing is driven by fibroblast proliferation, inducing substantial fibrous protein deposition, which ultimately allows the tendon to adhere to the surrounding tissue.^{51,64} Scar tissue, which is primarily caused by fibroblasts, is the root of epidural adhesions.⁶⁵ Therefore, inhibiting the proliferation of fibroblasts offers a potential strategy to prevent adhesions.

At the level of tissue state, postoperative tissue adhesions are closely associated with tissue healing, exhibiting distinct states. This process can be divided into three main parts. Firstly, the induction of an inflammatory response through the action of growth factors and cytokines. Secondly, the induction of tissue hypoxia, leading to the generation of reactive oxygen species and increased expression of vascular endothelial growth factor. Lastly, the reduced secretion of PAI, resulting in

diminished fibrinolysis and increased deposition of fibrous and collagenous tissues. These interconnected processes synergistically contribute to the overall adhesion formation.^{61,66} Excessive inflammatory response is a major contributor to tissue adhesion, with cytokines serving as key regulators of the inflammatory response.⁶⁷ Cytokines, including interleukins (ILs), tumor necrosis factor-alpha (TNF- α), and transforming growth factor-beta (TGF- β), are key factors in orchestrating the inflammatory response. TNF- α induces the production of IL-6, exacerbating inflammation and ECM synthesis. IL-1 and TNF- α further induce IL-6, contributing to adhesion formation. The overexpression of TGF- β triggers fibrosis and tissue adhesion. These cytokines act both individually and synergistically, and thus a profound understanding of the role of inflammation and its released mediators in adhesion formation is essential for the prevention of scarring and adhesion development.^{68,69} Besides inflammation, hypoxia and oxidative stress are inevitable pathways in tissue adhesion. Oxidative stress is the reaction of an organism to pathological oxidation or secondary injury, leading to the excessive generation of reactive oxygen species. Tissue cell damage during surgery induces hypoxia, initiating cascading intracellular reactions that intensify oxidative stress. A hypoxia-induced increase in free radicals leads to DNA and tissue damage, slowing down tissue healing and creating conditions for fibrosis and scar formation.⁶¹ Hypoxia also stimulates fibroblast proliferation, producing fibrotic factors, which exacerbate adhesion formation.^{70–72} An inflammatory response induced by oxidative stress increases the vascular permeability, releasing inflammatory factors such as TNF- α and ILs, which accelerate the formation of adhesions.^{73,74} Moreover, an imbalance in the fibrinolysis system contributes to tissue adhesion.⁶⁸ These tissue conditions (inflammation, hypoxia, oxidative stress, and fibrinolysis system imbalance) complement each other and are widely present in various tissue adhesions in the human body, collectively contributing to the formation of adhesions.

Understanding the mechanisms underlying various tissue adhesions is crucial for the design of anti-adhesion materials. One approach involves the modulation of cellular behavior by adjusting the physical and chemical properties of the materials (such as surface characteristics) to prevent adhesion. Another approach involves the incorporation of drugs or biological agents in materials to modulate both cellular behavior and tissue states, achieving enhanced anti-adhesion effects. In addition to regulating cellular behavior and tissue states, the indispensable characteristics for anti-adhesion materials include appropriate biocompatibility, tunable degradation rate, appropriate retention time, and outstanding mechanical properties.

In the following sections, we discuss the anti-adhesion materials used for various tissues. Some anti-adhesion materials involve modulating cellular behavior by preparing physical barriers with unique structures or properties. Others combine physical barriers with drugs capable of suppressing cell expression, exhibiting antimicrobial and antioxidant pro-

properties and regulating the fibrinolytic system to improve tissue states. Additionally, a portion of the research focuses on delivering bioactive substances through barriers to simulate the functionality of human tissues. These strategies comprehensively consider cellular behavior and tissue states, offering a more holistic solution for preventing tissue adhesion.

3. Electrospun nanofibrous membranes for tendon anti-adhesion

Tendons, as integral components of the musculoskeletal system, play a crucial role in facilitating physical movements and activities.^{75,76} However, they are prone to rupturing due to repetitive motion and degradation. Moreover, the unbalanced development of intrinsic healing and extrinsic healing often causes tissue adhesion during tendon repair, leading to restricted tendon movement and postoperative pain.^{77–79} Electrospun nanofibrous membranes have emerged as a promising materials for preventing tendon adhesions. Firstly, they function similarly to tendon sheaths, serving as physical barriers. Secondly, their high porosity and large surface area endow them with microstructural similarity to the ECM. Additionally, they offer adjustable structural parameters (fiber arrangement and diameter), mechanical properties, and degradation rates. Lastly, they can regulate local inflammatory responses through controlled drug or growth factor release.^{80,81} Anti-adhesive membranes prepared through electrospinning not only effectively reduce adhesions but also promote tendon repair.⁸² In this section, we classify tendon anti-adhesion membranes into two categories, *i.e.*, single-layer nanofibrous membranes and multi-layer nanofibrous membranes based on their hierarchical structure.

3.1. Single-layer nanofibrous membranes

The simplest single-layer nanofibrous anti-adhesive membranes can be obtained by directly electrospinning biodegradable polymers, which can achieve the performance of tendon anti-adhesion due to their physical barrier properties.⁸³ Meanwhile, electrospun nanofibers are also good drug loading platforms, enabling enhanced attributes of single-layer anti-adhesive membranes through drug integration.⁸⁴ In the following part, we focus on membranes with therapeutic agents and membranes without therapeutic agents.

3.1.1. Membranes with therapeutic agents. Research has indicated that the incorporation of therapeutic agents in NFMs can significantly enhance the overall anti-adhesive properties of the materials by modulating the healing microenvironment, thereby reducing the formation of adhesions.^{85,86} These therapeutic agents operate through diverse mechanisms, including the induction of fibroblast apoptosis, inhibition of DNA replication, transcription, and translation, anti-inflammatory and antioxidant activities, hemostasis, suppression of fibrous tissue-inducing growth factors, and modulation of the blood circulation, all aimed at preventing adhesion

formation.^{73,87} Depending on the formation mechanism of different tissue adhesions, we can choose the appropriate therapeutic agents for the preparation of anti-adhesion materials.

Tendon adhesion is a common complication following tendon injuries.⁸⁸ In this case, fibroblast migration, proliferation, and excessive collagen synthesis are recognized as important factors in adhesion development. Nowadays, various physical barrier anti-adhesion membranes have been developed to prevent the formation of tendon adhesions.^{89,90} However, besides fibroblast infiltration, inflammatory reactions also play a significant role in the development of tendon adhesions.⁹¹ Thus, fabricating nanofibrous membranes loaded with anti-inflammatory agents can greatly improve their anti-adhesion efficacies. Among the various therapeutic agents, non-steroidal anti-inflammatory drugs (NSAIDs) have attracted attention due their outstanding anti-inflammatory properties, which can prevent adhesion formation by blocking the mass migration of inflammatory cells.⁹² Celecoxib is a typical type of NSAID, which can inhibit fibroblast proliferation and collagen expression by inhibiting ERK1/2 and SMAD2/3 phosphorylation in cells.^{85,93} Therefore, Jiang and co-authors introduced the NSAID of celecoxib in poly(L-lactic acid)-polyethylene glycol (PELA) nanofibers *via* a blend-electrospinning technique to fabricate PELA/celecoxib composite nanofibrous anti-adhesive membranes, as shown in Fig. 2a.³² The obtained membranes with various celecoxib concentrations (2%, 6%, and 10%) were abbreviated as PELA-2%, PELA-6%, and PELA-10%, respectively. The results of *in vitro* drug release indicated that the burst release rates of the PELA-2%, PELA-6%, and PELA-10% membranes in the first two days were 32.8%, 40.7%, and 47.3%, respectively. Furthermore, after 20 days, nearly all the drugs were released from the fiber scaffolds of PELA-6% and PELA-10% (Fig. 2b). This observation suggests that the initial rate of drug release increases with a higher celecoxib concentration, while the overall duration of drug release remains nearly constant. Meanwhile, the area of fibroblasts growing on the surface of the composite membrane decreased with an increase in the celecoxib concentration (Fig. 2c), suggesting that the celecoxib-loaded PELA membrane could effectively prevent cell adhesion and provide an anti-tissue adhesion effect. However, its short drug release duration limits its potential for long-term drug delivery, which can hinder tendon healing.

Ibuprofen (IBU) is another commonly used NSAID, which can prevent adhesion formation by preventing the mass migration of inflammatory cells.⁹⁴ Cui and co-authors introduced IBU in poly(L-lactic acid) (PLLA) nanofibers to use in tendon anti-adhesion firstly.⁹⁵ IBU can inhibit macrophage adhesion, proliferation, and infiltration by blocking the expression of the Cox gene, and then suppress the expression of TNF- α and collagen III deposition. Thereby, it alleviates inflammation and prevents tendon adhesion. Ingeniously, to prevent the occurrence of secondary adhesion caused by the degradation of the fibers after the complete release of IBU, they loaded IBU into modified mesoporous silica (MMS) nano-

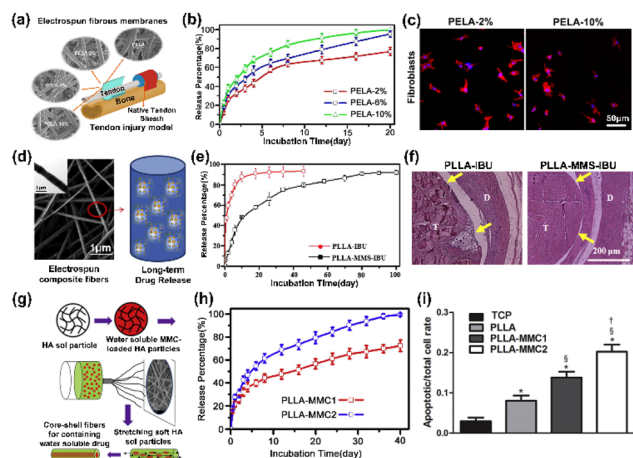


Fig. 2 Composition, structure, and performance of single-layer nanofibrous membranes with therapeutic agents. (a) Morphology of celecoxib-loaded PELA nanofibrous membrane. (b) Cumulative celecoxib release from nanofibers after incubation in PBS at 37 °C. (c) Fluorescent micrographs of dermal fibroblasts after 24 h of culture on PELA-2% and PELA-10% membranes. Reprinted with permission from ref. 32. Copyright 2014, Elsevier. (d) Microstructure and schematic of electrospun composite nanofibers encapsulated with IBU-loaded MMS. (e) *In vitro* IBU release profiles of PLLA-IBU and PLLA-MMS-IBU nanofibers. The samples were incubated in PBS at 37 °C. (f) H&E staining of tissue sections after 8 weeks. Repair site encircled with PLLA-IBU membrane and PLLA-MSN-IBU membrane, respectively. D indicates dense connective tissue, T indicates surrounding the tendon, while yellow arrowheads indicate the interface without adhesion tissue. Reprinted with permission from ref. 95. Copyright 2013, Elsevier. (g) Schematic illustrations of preparation and structure of MMC-loaded PLLA nanofibers by micro-sol electrospinning. (h) Cumulative release of MMC from PLLA-MMC fibers after incubation in PBS at 37 °C. (i) Quantification of cell apoptosis ratio of NIH/3T3 fibroblasts on the surfaces of different samples after 24 h of culture. Reprinted with permission from ref. 104. Copyright 2015, Elsevier.

particles firstly, and then encapsulated them into nanofibers using a co-solvent electrospinning method to achieve long-term drug release (more than 100 days). Benefiting from this novel composite release system, a single-layer nanofiber tendon anti-adhesion membrane (PLLA-MMS-IBU) with dual slow-release capabilities was successfully constructed. The IBU-loaded MMS nanoparticles were evenly distributed in the nanofibers (Fig. 2d). Compared with the conventional PLLA-IBU nanofiber membranes, the PLLA-MMS-IBU membranes demonstrated a significantly reduced initial burst release (only 6% within the initial 12 h, Fig. 2e) due to their dual slow-release effects. Moreover, the control PLLA-IBU group had completely released the drug after approximately 20 days, while the PLLA-MMS-IBU membranes released around 91% of the drug by 100 days. Additionally, the results of histological analysis of postoperative peritendinous adhesions and tendon healing indicated that the PLLA-MMS-IBU group showed lower tissue adhesion at 8 weeks than that of the PLLA-IBU group (Fig. 2f). This study provides a simple strategy for constructing nanofiber platforms with long-term sustained drug release characteristics, which may function as a novel

carrier for long-term anti-inflammation and anti-adhesion to prevent peritendinous adhesions. However, a fatal disadvantage of these membranes should be considered, namely, nanoparticles are prone to fall off during use, posing a huge potential risk to the body and tendon healing.^{96,97}

With the deepening of research, some studies have confirmed that inducing fibroblast apoptosis and inhibiting collagen secretion is a superior method to prevent adhesion formation from the perspective of tendon healing and adhesion formation mechanism.^{98–100} Also, mitomycin C (MMC) is an ideal candidate to achieve these aims.^{101,102} MMC has been demonstrated to induce apoptosis in fibroblasts by regulating miR-200b and RhoE.¹⁰³ Therefore, the same research group, Cui and co-authors, loaded MMC in hyaluronic acid (HA) hydrosol nanoparticles firstly, and then encapsulated them into PLLA nanofibers *via* micro-sol electrospinning.¹⁰⁴ This approach neatly solved the problem that hydrophilic MMC has a low drug encapsulation efficiency in hydrophobic PLLA fibers. Moreover, the prepared nanofibers exhibited a typical core-shell structure (PLLA sheath and MMC/HA core) due to the microphase separation, which can encapsulate the drug in the center of the fiber and extend the release time (almost completely released within 40 days), as illustrated in Fig. 2g and h. Additionally, introducing MMC in PLLA nanofibers not only inhibits the extrinsic healing of tendons but also promotes intrinsic healing. The apoptotic rates of fibroblasts on the surfaces of the different samples revealed that the membranes loaded with MMC could effectively reduce the cell viability, thereby preventing fibroblast proliferation and adhesion (Fig. 2i). These materials achieved the prevention of tendon adhesion formation; meanwhile, they had a great effect on intrinsic tendon healing, avoiding drug burst release simultaneously. Nevertheless, the use of MMC is complicated with much acute and chronic toxicity, which should not be ignored.

Furthermore, research suggests that the activation of the nuclear factor-kappa B (NF- κ B) pathway in macrophages is a crucial factor in the excessive synthesis of collagen and the formation of tendon adhesions.¹⁰⁵ Lu *et al.* incorporated pyrrolidine dithiocarbamate (PDTTC) in electrospun nanofibrous membranes of poly (L-lactic acid) (PLA) using the electrospinning technique for preventing tendon adhesions.¹⁰⁶ PDTTC is a targeted NF- κ B inhibitor capable of permeating cell membranes. The results indicated that the electrospun nanofibrous membranes loaded with PDTTC could inhibit NF- κ B activation, reduce the release of pro-inflammatory factors, control inflammation, and thereby enhance the anti-adhesion effect. This finding has prompted further investigations into the potential advantages of NF- κ B inhibition for anti-adhesion purposes in the long term.

3.1.2. Membranes without therapeutic agents. Although loading anti-inflammatory or cell expression-inhibiting drugs into electrospun nanofibers has shown high efficacy in preventing adhesion formation, the inherent local side effects of these drugs inevitably compromise the growth of normal cells, thus retarding the tendon healing process.¹⁰⁷ Numerous studies have demonstrated that superhydrophobic or superlu-

bricated surfaces have excellent anti-cell adhesion effects.^{108–110} Therefore, the development of electrospun nanofibers with unique surface interfaces can effectively solve the above-mentioned problems.

The mixing method is the easiest way to endow to electrospun nanofibrous membranes with super hydrophobicity, effectively resisting the non-specific adsorption of cells and proteins. Therefore, Liu and co-authors incorporated naturally hydrophobic polymer beeswax (Wax) into PLA solutions to prepare Wax/PLA nanofiber tendon anti-adhesion membranes by using the blending electrospinning technique.¹⁰⁸ The obtained bi-component nanofibers featured uniform diameters and a super-hydrophobic surface (contact angle of 152°), as depicted in Fig. 3a and b, respectively. The *in vitro* cell adhesion experiments results revealed that the proliferation area of fibroblasts on the Wax/PLA membranes was considerably less than that on PLA membranes, further indicating that the surface hydrophobicity of the electrospun nanofibers plays a crucial role in reducing the adhesion and proliferation of fibroblasts (Fig. 3c). However, the variety of superhydrophobic polymers with excellent biodegradability that can be used for electrospinning is limited, which imposes great restriction on the further development of this method.

Besides superhydrophobic surfaces, researchers have found that superlubricated surfaces are also effective in preventing cell adhesion.¹¹¹ The construction of a stable hydration lubrication layer on the surface of nanofibers is a promising approach to achieve superlubrication, which can not only

create a low friction coefficient (COF) between two surfaces under shear, but also enable easy detachment of adhered proteins from the material surface.^{112,113} On these highly lubricated surfaces, cellular pseudopods are difficult to form robust focal adhesion, and thereby inhibition of cell adhesion can be achieved. Therefore, based on this principle, Cheng and co-authors created poly(ϵ -caprolactone) (PCL) nanofibers with superlubricated surfaces *via* the one-step *in situ* grafting of a copolymer synthesized using dopamine methacrylamide and 2-methacryloyloxyethyl phosphorylcholine (pMPC) and the dip-coating method, as illustrated in Fig. 3d.¹¹⁰ The obtained membranes exhibited a favorable lubrication performance, and the optimal COF of this nanofibrous membrane was as low as 0.1, which is approximately 65% lower than that (0.32) of the pure PCL nanofibers (Fig. 3e). Furthermore, the *in vivo* experiments also demonstrated that the membranes with lubricating surfaces exhibited lower adhesion scores and enhanced tissue anti-adhesion effect, without sacrificing intrinsic tendon healing. However, this method involves a complex post-treatment process of dip-coating, which limits the large-scale preparation.

Subsequently, Zhang and co-authors built a super-lubricated surface on a PCL/poly (2-methacryloyloxyethyl phosphorylcholine) (PMPC) composite nanofibrous membrane *via* a simple blend-electrospinning by controlling the relative humidity (RH), as shown in Fig. 3f.¹⁰⁹ This approach ensured both the thorough phase separation of PMPC during electrospinning and a high density of zwitterionic charges on the nanofiber surface. In particular, the COF of the composite nanofibers containing 10% PMPC and produced at 65% RH was less than 0.05, which is similar to that of the surface of ice (Fig. 3g). Additionally, the *in vitro* cell adhesion data revealed that the composite nanofiber group possessed strong resistance to fibroblast adhesion compared to the pure PCL nanofiber group. Moreover, the cell proliferation became nearly negligible when the PMPC content exceeded 10% (Fig. 3h). This suggests that the use of nanofiber membranes with a superlubricated surface for tendon anti-adhesion is very promising. However, although the anti-adhesion properties of these materials can be substantially improved by endowing them with hydration lubrication, the preparation method is sensitive to the environmental humidity, and the property stability of this material needs to be improved.

3.2. Multi-layer nanofibrous membranes

The tendon sheath is an important structure encircling the exterior of the tendon, which consists of an outer fibrous sheath and an inner synovial sheath. The fibrous layer and synovial layer form a closed channel, within which synovial fluids (primarily composed of HA) provide lubrication and nourishment to the tendon.^{75,76} The outer fibrous layer, serving as a robust biological barrier, prevents tendon adhesion caused by the infiltration of surrounding fibrous tissues.⁸² Accordingly, creating a tendon sheath using multi-layer electrospun nanofibrous membranes is a novel idea to inhibit adhesion and enhance tendon repair.¹¹⁴ Currently,

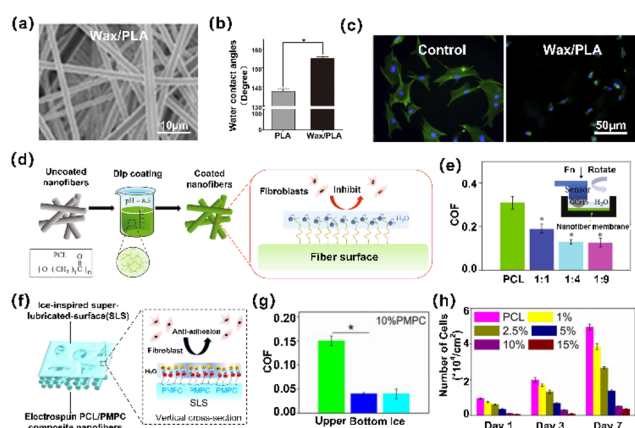


Fig. 3 Composition, surface property, and performance of single-layer nanofibrous membranes without therapeutic agents. (a) SEM image of Wax/PLA nanofibers. (b) Water contact angles of PLA and Wax/PLA nanofibers. *indicates $P < 0.05$. (c) Cellular arrangement area of merge surface at day 4. Reprinted with permission from ref. 108. Copyright 2020, Elsevier. (d) Dip-coating method to graft pMPC onto the PCL fibrous membrane surface. (e) COF of all the samples. Inset is a schematic diagram of the rotation mode. Reprinted with permission from ref. 110. Copyright 2020, American Association for the Advancement of Science. (f) Illustration of super-lubricated surface of PCL/PMPC nanofibers. (g) COF of upper and bottom surfaces of 65% RH samples and ice surface. (h) Average number of cells measured from the confocal laser scanning microscopy images. Reprinted with permission from ref. 142. Copyright 2020, the American Chemical Society.

much attention has been paid to the development of multi-layer biomimetic nanofibrous membranes for preventing tendon adhesions. In the following sections, the discussion will focus on multi-layer nanofibrous membranes with and without therapeutic agents.

3.2.1. Membranes with therapeutic agents. Introducing anti-adhesive materials into the human body often triggers inflammatory responses, which can reduce their anti-adhesive efficacy. Thus, to weaken the inflammatory reactions and promote tendon healing, Fan and co-authors designed a multi-functional biomimetic tendon sheath composed of multi-layer membranes (Fig. 4a and b).¹¹⁴ Celecoxib-loaded PELA nanofibers served as the outer layer and pure PELA nanofibers acted as the inner layer. Meanwhile, between these two layers, an HA gel layer was present. The celecoxib-loaded nanofibers layer mimics the outer layer of the natural tendon sheath, which plays an important role in resisting cellular invasion by inhibiting fibroblast growth and collagen synthesis. The intermediate gel layer is responsible for releasing HA synovial fluid to lubricate and nourish the tendon. Alternatively, the PELA nanofiber inner layer is a substitute for the natural tendon sheath in the synovial layer, which promotes tendon gliding.

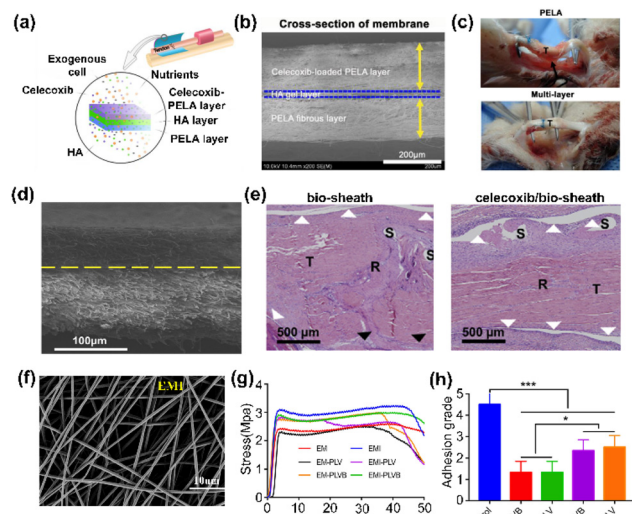


Fig. 4 Multi-layer biomimetic nanofibrous anti-adhesion membranes with therapeutic agents. (a) Schematic illustration of multi-layer electrospun membrane. (b) SEM image of cross-sectional features of the multi-layer membrane. (c) Gross evaluation of the rabbit flexor digitorum profundus tendon model in PELA membrane group and multi-layer membrane group. Tendon (T) is indicated in the figures and adhesion tissue is indicated by black arrows. Reprinted with permission from ref. 114. Copyright 2015, Multidisciplinary Digital Publishing Institute. (d) Cross-section of celecoxib/bio-sheath nanofiber membrane. (e) HE staining of repair site wrapped with bio-sheath and repair site wrapped with celecoxib/bio-sheath. Black arrowheads indicate adhesion tissue surrounding the tendon (T), while white arrowheads indicate the interface without adhesion tissue. Sutured sites (S) are indicated in the pictures. Reprinted with permission from ref. 115. Copyright 2016, Elsevier. (f) SEM image of EMI. (g) Stress-strain curves of the composite membrane. (h) Tendon adhesion score (general observation). Reprinted with permission from ref. 116. Copyright 2021, Elsevier.

The results of *in vivo* animal experiments showed that adhesion still occurred in the tendons of mice using an ordinary PELA nanofiber membrane, whereas there was almost no adhesion around the tendons repaired with the multi-layer membrane group (Fig. 4c). This multifunctional drug-loaded biomimetic tendon sheath membrane not only enhanced the anti-adhesion effects but also facilitated tissue healing simultaneously. However, the bending resistance of these membranes are weak due to the poor mechanical properties of the HA gel layer and the interface bonding between the gel layer and the fiber layers needs to be further improved.

Subsequently, Ruan and co-authors prepared a celecoxib-containing bilayer bionic tendon sheath *via* a continuous and microgel electrospinning technique.¹¹⁵ The outer layer of the celecoxib-containing PELA electrospun membrane acted as a substitute for the fiber layer in natural tendon sheath and the inner layer composed of HA/PELA core-sheath nanofibers mimicked the synovial layer (Fig. 4d). This dual-layer membrane design not only better mimicked the natural tendon sheath but also subtly avoided the defect of poor binding force between the gel and fiber layers. The histological assessment of the tendon repaired indicated that there was only a small amount of adherent tissue in tendon wrapped by the bionic tendon sheath, while the control group had severe adhesions (Fig. 4e). In addition, the bionic tendon sheath containing celecoxib showed the best anti-adhesion effect, with no adherent tissue and a clear tendon sheath gap in the HE-stained sections.

Basic fibroblast growth factor (bFGF) is a good protein drug that can promote tendon healing, but the bioactivity of free bFGF is easily inactivated. Therefore, Xiao and co-authors encapsulated bFGF in a poly(ethylene glycol)-*block*-poly(L-valine) (PEG-PLV) hydrogel (named PLVB) on the surface of a poly(lactic-co-glycolic) acid (PLGA) electrospun nanofibrous membrane loaded with IBU (named EMI) to prepare a double-layer composite membrane named EMI-PLVB.¹¹⁶ IBU plays a significant antimicrobial and anti-inflammatory role to inhibit exogenous healing, while bFGF accelerates tendon healing. Moreover, the EMI layer shows a porous structure, which allows nutrients from the tissue to be delivered to the tendon injury site, contributing to promoting endogenous healing. Also, the composite of hydrogel and electrospun membrane greatly improved their biodegradability and stability (Fig. 4g). In addition, the tendon surface at the tendon repair site in the EMI-PLVB group was smooth and showed the lowest adhesion score (Fig. 4h).

3.2.2. Membranes without therapeutic agents. Although drug-loaded nanofiber membranes can achieve a relatively good anti-adhesion effect, there are many drawbacks, such as the uncontrolled drug load, unknown drug interactions, and complicated preparation process.¹¹⁷ Therefore, research on the use of material structure or the addition of biological agents to prevent adhesion is booming. Qian and co-authors developed an innovative dual-layer anti-adhesion membrane by integrating two PLLA electrospun nanofiber layers into a single layer using shear force, as shown in Fig. 5a.¹¹⁸ The purpose of the

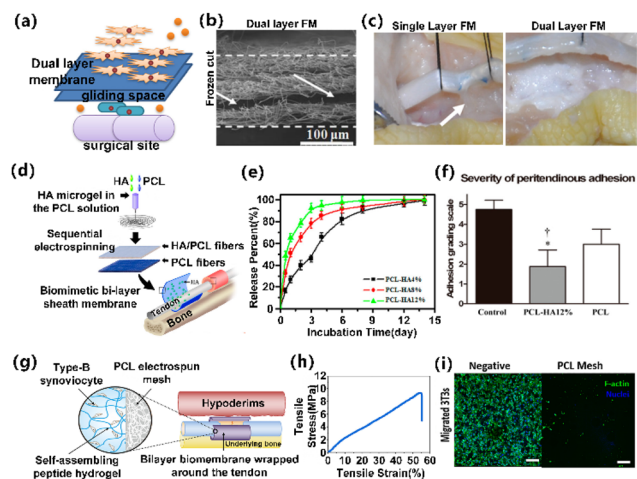


Fig. 5 Multi-layer biomimetic nanofibrous anti-adhesion membranes without therapeutic agents. (a) Illustration of the mechanism of the dual-layer PLLA anti-adhesion membrane. (b) Cross-sectional view of PLLA nanofibrous membranes. (c) Gross evaluation of tendon repair in chicken models of flexor digitorum profundus tendon injury. Reprinted with permission from ref. 118. Copyright 2020, Frontiers. (d) Schematic of fabricating biomimetic bilayer sheath membrane. (e) *In vitro* HA release cumulative percentage from electrospun fibers with drug loadings of 4%, 8%, and 12% after incubation in PBS at 37 °C. (f) Adhesion was graded according to severity of peritendinous adhesion. Reprinted with permission from ref. 119. Copyright 2012, the American Chemical Society. (g) Schematic of proposed approach to impede adhesion formations. A bilayer biomembrane is wrapped around the tendon. (h) Representative stress-strain curve under tension, showing the predominantly elastic behavior of the PCL mesh. (i) Immunofluorescence of 3T3 cells migrated through the PCL mesh compared to negative control (8 μm pore insert membrane) shows minimal fibroblast migration. Migrated cells were stained for F-actin (green) and nuclei (blue). Scale bar = 100 μm. Reprinted with permission from ref. 120. Copyright 2021, Elsevier.

bilayer design is to promote tendon sliding, while acting as an anti-adhesion agent. There are some spaces between the two layers, which are favorable for tendon gliding and inhibiting cell penetration (Fig. 5b). Moreover, the porous structure of the nanofibers allows signaling molecules and nutrients to be exchanged through the membrane, increasing collagen levels and stimulating tendon regeneration. In the *in vitro* experiments in animals, adhesions between the regenerated tendon tissue and the surrounding tissues were seen in the group repaired using a single membrane, while the site repaired in the bilayer membrane group was smoother and free of adhesions (Fig. 5c). This suggests that it is feasible to use the bilayer membrane group without drug delivery to prevent tendon adhesions, which lays the foundation for further research and clinical translation in the future. However, this anti-tendon adhesion membrane cannot secrete synovial fluid like natural tendon sheaths and does not serve to nourish the tendon.

Liu and co-authors prepared a bilayer bionic tendon sheath consisting of an HA-loaded PCL fibrous membrane as the inner layer and PCL fibrous membrane as the outer layer by a

combination of sequential and microgel electrospinning technologies (Fig. 5d).¹¹⁹ The inner PCL-HA fiber layer mimics the release of synovial fluid from the tendon sheath, while the outer layer prevents cell adhesions. The *in vitro* HA release results showed that a distinct burst release phase, followed by a gradual release occurred (Fig. 5e). Also, benefiting from the advantages of electrospun fibrous scaffolds, the HA release time could last up to 14 days. Moreover, the average scores in the biomimetic bilayer sheath membrane group were lower than the other groups ($P < 0.05$; Fig. 5f) based on the severity of peritendinous adhesion scores from macroscopic observations. Further animal studies showed that the outer sheath membrane impeded adhesion to the surrounding tissues, whereas the inner HA-PCL layer facilitated the sliding and healing process of the tendon.

The preparation of ideal scaffold materials for the repair and regeneration of damaged tissues to fully restore the function of the original tissue is of great importance in the field of tissue engineering. However, scaffolds prepared based on biomaterials may not be able to properly regulate the cellular behavior of tissue assembly. Therefore, the development of biomaterial scaffolds containing biological agents such as cells and growth factors has received significant attention. There are many tendon anti-adhesion membranes containing biological agents, such as HA membranes and human amniotic membranes. Among them, HA-based compounds are more widely used, but these natural polymers have a common problem in that they are easily degraded *in vivo* and have limited trophic function for tendons, which cannot restore the structure of the synovial sheath, resulting in the loss of their barrier function.

Given this, Angela Imere and co-authors constructed a biologically active bilayer biofilm by combining the electrospinning and self-assemble methods.¹²⁰ The outer layer of this biofilm was a PCL nanofibrous membrane and the inner layer was self-assembled peptide hydrogels (SAPHs) containing synovial B cells, as shown in Fig. 5g.^{53,121,122} This bilayer membrane acted as an anti-adhesion system capable of restoring tendon lubrication, while assisting synovial sheath regeneration. The SAPH layer exhibited good lubrication properties due to the loading of synovial B cells, which can realize the sustained endogenous production of HA. Moreover, the SAPH layer was conducive to synovioyte growth and proliferation, which can promote synovial sheath repair and tendon sliding. The outer PCL nanofiber layer acted as a physical barrier to prevent fibroblast proliferation and ensured the overall mechanical properties of the membrane. As shown in Fig. 5h, the PCL membrane exhibited predominantly elastic behavior under tensile and finite plastic deformation, which ensured easy fixation of the product during implantation without the risk of fracture. In addition, immunofluorescence analysis showed that it is difficult for fibroblasts to proliferate on the PCL nanofiber membranes, demonstrating good ability of inhibiting cell growth (Fig. 5i). Therefore, this material can not only be used as a physical barrier against cell adhesion but also produce HA to restore the sliding performance and func-

tion of the tendon in the long term, showing great potential in clinical practice.

Moreover, gene-based therapeutics have been devised for the prevention of tendon adhesion. The objective of gene delivery is to introduce exogenous genetic material into cells to alter DNA, inducing, silencing, upregulating, or downregulating the expression profile and secretion of proteins, thereby achieving the goal of anti-adhesion. In recent years, there has been progress in the development of gene silencing through small interfering RNA (siRNA) to downregulate specific genes. Liu and colleagues developed a cationic 2,6-pyridinedicarboxaldehyde-polyethylenimine (PDA)-mediated extracellular signal-regulated kinase (ERK)2-siRNA polymeric delivery system, which was electrospun into PLLA/HA nanofibers.¹²⁷ The results demonstrated that this composite membrane can attenuate cell proliferation, restrict the deposition of collagen III, and downregulate the gene expression of ERK2 and its downstream mediator SMAD3. The advantage of this method lies in its ability to selectively modulate key cellular signals in the adhesion mechanism and relevant downstream pathways, effectively inhibiting adhesion formation.

4. Electrospun nanofibrous membranes for abdominal anti-adhesion

Abdominal adhesions are one of the more common complications after abdominal surgery, with an incidence as high as 90% to 95%.^{53,121,122} The results of a large number of clinical trials have shown that the formation of adhesions is associated with trauma, infection, and foreign body reaction.¹²³ Undesirable adhesions often lead to serious complications, including chronic pain, intestinal obstruction, infertility in women, and the need for a second surgical intervention, which poses a great threat to human health.^{124–126}

Thus, preventing abdominal adhesion by using electrospun nanofibrous membranes has been a research hotspot application due to their integrated characteristics of tunable surface properties, controlled porous structure, and significant surface area-to-volume ratio.^{128,129} In this section, we categorize electrospun nanofibrous membranes into two categories based on their distinct applications in preventing abdominal adhesion, *i.e.*, membranes as peritoneal anti-adhesion materials and membranes as hernia repair materials.

4.1. Nanofibrous membranes as peritoneal anti-adhesion materials

A typical type of postoperative abdominal adhesion is peritoneal adhesion, which refers to the formation of a pathologic connection between the injury site and the peritoneum.¹³⁰ Improving the surface hydrophobicity of electrospun nanofibrous membranes is a feasible and commonly used method to optimize their anti-adhesion capacities. Based on this principle, Klicova and co-authors electrospun PCL nanodroplets

(ND) on the surface of pre-prepared PCL nanofibers (NF) to form a multilevel-structured anti-adhesion membrane inspired by lotus leaf nanostructures, as shown in Fig. 6a and b.¹³¹ Then, the surface hydrophobicity of the membrane surface was further enhanced through plasma modification. The water contact angle could be increased from 99° to 125° by extending the plasma treatment time to 3 min (Fig. 6c). The results of *in vitro* experiments showed that the proliferation of fibroblasts on the hydrophobic surface decreased dramatically. This nanofibrous membrane affected cellular behavior by modulating the structural properties of its surface, inhibiting cell adhesion and proliferation. Besides, the results of cytotoxicity experiments showed that the plasma-treated nanofibrous membranes were non-cytotoxic. However, although this method works well against peritoneal adhesion, it cannot mimic the structure of the natural peritoneum.

The natural peritoneum is a smooth, transparent plasma membrane consisting of a mesothelial layer and a connective tissue layer.¹³² The mesothelial layer secretes surfactants and glycosaminoglycans for lubrication, which contributes to frictionless sliding of organs in the peritoneal cavity.¹³³ To mimic the structure of the peritoneal membrane, nanofibrous membranes can be lubricated by enhancing the surface wettability

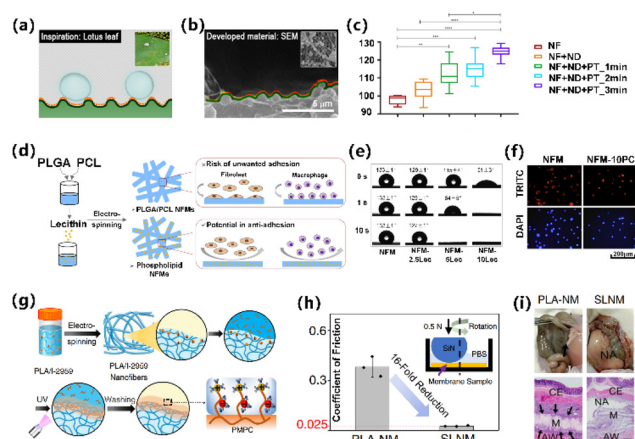


Fig. 6 Some representative electrospun nanofibers for peritoneal anti-adhesion. (a) Schematic illustration of the lotus leaf structure. (b) SEM of side view of the produced material. (c) Contact angles of tested materials. Reprinted with permission from ref. 131. Copyright 2022, Elsevier. (d) Schematic of method for the preparation of phospholipid-functionalized nanofiber-based barrier and its potential in anti-adhesion. (e) Contact angles of diverse NFMs at different times. (f) Morphologies of L-929 fibroblasts after 4 h of seeding on NFM and NFM-10PC. Reprinted with permission from ref. 134. Copyright 2021, Elsevier. (g) Schematic diagram of optimized method of subsurface-initiated polymerization. (h) Comparison of COF value between the PLA-NM and SLNM samples based on the tribological test operated under a rotation mode ($n = 3$ independent experiments). PLA-NM PLA nanofibrous membrane and SLNM nanofibers with superlubricated nanoskin. (i) Photos of the harvested cecum and abdominal wall and H&E staining as well as Masson staining images of the harvested tissues on day 7. The black arrows point to adhesion site. CE cecum. M membrane. AW abdominal wall. NA no adhesion. Reprinted with permission from ref. 135. Copyright 2020, Springer Nature.

to prepare a multifunctional barrier. Phospholipids, a common amphiphilic substance occurring naturally in the peritoneal cavity with excellent lubricating properties, have been demonstrated significant prevention of postoperative adhesion without any adverse effects. Therefore, Wang and co-authors doped lecithin, a natural combination of phospholipids, in PLGA/PCL nanofibers *via* a one-step electrospinning technique to construct phospholipid-functionalized nanofiber-based barriers with superior wettability, which could form a lubricating layer to resist unwanted adhesions (Fig. 6d).¹³⁴ The surface wettability of the membranes shifted from highly hydrophobic to extremely hydrophilic after the introduction of lecithin and the time for the contact angle to decrease to 0° was shortened from 10 s to 1 s when the lecithin concentration increased from 5% to 10% (Fig. 6e). Due to the superhydrophilic property of the membranes, a hydrated layer with lubrication effect could be formed on their surface, endowing them with anti-adhesion performance. The fibroblast adhesion results showed a significant reduction in the cells on the surface of the phospholipid-functionalized nanofibrous membranes, indicating a good efficacy of preventing adhesion formation. However, the hydration of this material was relatively limited due to the uniform distribution of lecithin in the fibers.

The construction of a robustly bonded, fully covered, and superlubricated coating on the surface of electrospun nanofibers is an excellent strategy to achieve the prevention of postoperative adhesion. Therefore, Wang and co-authors generated a layer of superlubricated nanoskin on the surface of a single nanofiber using a zwitterionic polymer and an inside-out method.¹³⁵ The initiator 2-hydroxy-1-[4-(2-hydroxyethoxy)phenyl]-2-methyl-1-propanone was added to a PLA solution and electrospun into nanofibers. Then, the nanofibrous membranes were immersed in an aqueous solution of MPC monomer and exposed to ultraviolet light to complete photopolymerization and generate PMPC nanoskin on the surface of single nanofibers (Fig. 6g). This facile approach of subsurface-initiated inside-out growth of *in situ* superlubricated nano-skin on electrospun nanofiber surface allowed the nanoskin to be generated in the superficial layer of the nanofiber, which has the advantages of strong binding, full coverage, and nanoscale encapsulation. The authors denoted the prepared nanoskin as SLNM. This unique design enabled the nanofibrous membrane to have ultra-low friction and almost 0° water contact angle, as shown in Fig. 6h. The average COF of the SLNM was about 0.025, which is only one-sixteenth of that of the PLA-NM. After 12 000 test cycles, the time-friction coefficient curve of SLNM remained stable, and no wear was found on the surface of the nanofibrous membrane, which indicates that the superlubrication performance of SLNM is very durable. The further animal experiment results showed the absence of adhesion in the SLNM group, while severe tissue adhesion occurred in the PLA-NM group (Fig. 6i). This is consistent with the H&E staining results, indicating that the SLNM group samples had the best anti-adhesion properties among the different materials tested. Therefore, the prepared super-lubricating electrospun nanofiber membrane exhibited an out-

standing performance in preventing postoperative tissue adhesion with the ability to influence cellular behavior to some extent.

Overall, the preparation of peritoneal anti-adhesion membranes using electrospinning technology has advantages that are incomparable to traditional methods. However, there are still many problems and challenges in this current research field. The most notable is that most of the studies are based on animal model experiments, which have large discrepancies between them and human experiments. In addition, the safety of these materials when applied to humans is an issue that needs to be pondered.

4.2. Nanofibrous membranes as hernia repair materials

The abdominal wall hernia is a common condition with a global prevalence of 4%–8% per year.^{136,137} Reports have shown that more than 1 million abdominal wall hernia repairs are performed in the United States annually, accounting for nearly \$2.5 billion in overall healthcare expenditures each year.^{138–140} In clinical practice, the repair of abdominal wall hernias with mesh has become the most common treatment. There are more than 100 types of surgical mesh available on the market, such as PP mesh and expanded polytetrafluoroethylene. Among them, PP mesh is the most commonly used hernia repair material, which has good chemical stability and mechanical properties and can provide support for human tissues to achieve therapeutic effects. However, pure PP mesh has no anti-adhesion effect, and its non-degradability can cause a foreign body reaction, ultimately leading to severe abdominal adhesions.^{141–145} Therefore, it is necessary to combine PP mesh with a physical barrier membrane to endow PP mesh with anti-adhesion properties or develop new hernia repair materials.^{146–153} Here, we focus on composite mesh combined with electrospun nanofibrous membranes and advanced biodegradable nanofibrous hernia repair materials.

Martin Plencner and co-authors loaded electrospun PCL nanofibrous membranes onto PP mesh, and then adsorbed growth factors (bFGF, TGF-β2) onto the surface of the composite mesh by impregnation to create a hernia repair mesh with significantly improved anti-adhesion and cell regeneration ability.¹⁵⁴ However, due to the non-degradable nature of the PP mesh, implantation of the PP mesh into the human body after hernia repair will inevitably cause an inflammatory reaction, which is detrimental to tissue reconstruction. Therefore, combining drugs with immune-suppressing effects and physical barriers to prevent adhesions will be more effective. Subsequently, Zhang and co-authors added sirolimus (SLR) to PLA nanofibers, and then combined the drug-loaded nanofibrous membranes with PP mesh to obtain a composite PP/PLA/SRL mesh, which was used to reduce the inflammatory reaction and promote the reconstruction of abdominal wall defects in rats.¹⁵⁵ In this study, the composite of drug-loaded nanofibrous membrane and PP mesh not only inhibited adhesion formation but also reduced the inflammatory response, which is beneficial for the reconstruction of the abdominal wall. However, SRL has an inhibitory effect on cell

proliferation, which was detrimental to tissue repair. Therefore, although the method of loading drugs using composite mesh is effective, there are still drawbacks that need to be further explored and investigated. Chitosan (CS), a deacetylated product of chitin, has excellent properties such as antimicrobial, biocompatibility, and biodegradability, making it a good choice in the biomedical field. A powerful antimicrobial composite mesh can be obtained by modifying PP mesh with the introduction of CS. Umran Aydemir Sezer and co-authors combined CS-loaded PLGA nanofibers with PP mesh for adhesion prevention after hernia surgery.¹⁴⁹ It was shown that the results of the composite mesh combined with CS were significantly better than the other groups in terms of lubrication properties, antimicrobial and hemostatic efficiency. Importantly, it not only showed a good anti-adhesion effect, but also circumvented the risk of cytotoxicity, as well as possessed some antimicrobial effect to maximize the prevention of adhesion.

However, although the anti-adhesion performance of PP mesh can be improved by forming a composite with nanofibers, a significant drawback to PP is its non-degradability, which easily causes a foreign body reaction after implantation. Then, Biopatch including decellularized dermal matrix and small intestinal submucosa was developed for the repair of abdominal wall defects. However, its poor mechanical strength usually leads to hernia recurrence. Reconstructing the subcutaneous tissue and the mural peritoneum in the abdominal wall is also a significant strategy to repair abdominal wall defects and restore their function, playing important roles in resisting intra-abdominal pressure to prevent the development of hernia and adhesions, respectively. In this case, bilayer patches that can mimic the function of the abdominal wall may be able to achieve good results in abdominal wall repair. Therefore, Liu and co-authors prepared a double-layer structured nanofibrous membrane with good anti-hernia and anti-adhesion effects by combining electrospinning and immersion-crosslinking methods (Fig. 7a).¹⁵⁶ A blend solution containing PCL and graphene oxide (GO) was first used to generate the outer scaffold and the inner layer composed of PCL and CS was directly spun onto the surface of the GO-PCL, and then the double-layered scaffold (GO-PCL/CS-PCL) was immersed into a mixture including *N*-acetylcysteine (NAC) to expand the bio-functions of the patch. The obtained composite GO-PCL/NAC-CS-PCL membranes showed a clear fiber morphology and obvious double-layer structure (Fig. 7b). The incorporation of GO in the outer layer improved the mechanical strength of the scaffolds, which contributed to avoiding hernia formation. Moreover, the CS in the inner layer can reduce local inflammatory reactions, and thus prevent the formation of adhesions due to its hemostatic and antimicrobial abilities. Furthermore, NAC has outstanding capability of promoting the repair of abdominal wall defects, which is ascribed to its great role of stimulating neovascularization and reducing the levels of reactive oxygen species at the site of injury.^{157,158} Thus, the scaffolds containing CS and NAC showed a substantial decrease in the adhesion fraction compared to the normal

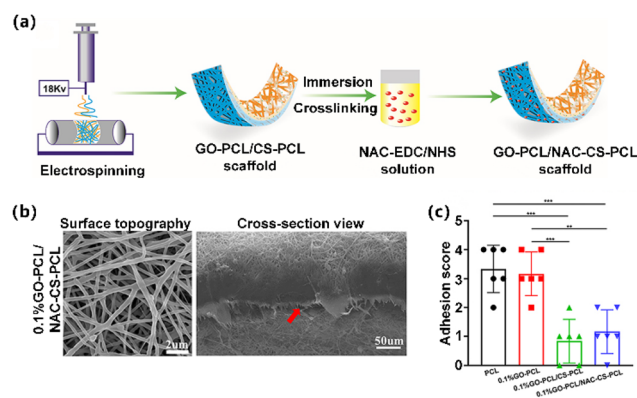


Fig. 7 Some representative electrospun nanofibers for hernia repair. (a) Illustration of the design and fabrication process of the GO-PCL/NAC-CS-PCL scaffold. (b) SEM images showing the surface and cross-section views of the scaffolds. The red arrow shows the double-layer structure. (c) Histogram of adhesion score (mean $\pm$ SD; ** P < 0.01, *** P < 0.001). Reprinted with permission from ref. 156. Copyright 2021, Dove.

scaffolds (Fig. 7c). Overall, using electrospun nanofibrous membranes with biodegradability instead of conventional PP hernia repair materials provides a promising platform as a postoperative anti-adhesion agent. However, endowing nanofibrous membranes with strong and stable mechanical properties is also a research direction that is worth exploring in the future.

5. Electrospun nanofibrous membranes for other postoperative anti-adhesions

5.1. Nanofibrous membranes for prevention of epidural adhesion

Epidural adhesion is an unavoidable complication after laminectomy, which usually leads to nerve compression, severe pain, and other clinical symptoms.^{159–161} Epidural fibrous protrusions and scar formation are the main causes of epidural adhesions. Additionally, collagen synthesis, accumulation of inflammatory cytokines, and proliferation of fibroblasts accelerate the formation of adhesions in the epidural operative area after surgical trauma.¹⁶² Therefore, minimizing epidural adhesions is a top priority.

Using anti-adhesion drugs is an effective and simple way to prevent postoperative tissue adhesion. To extend the release of the drug and achieve physical barrier effect, Huang and co-authors loaded Icaritin (ICA), a natural anti-adhesion drug, into PCL/gelatin (GT) nanofibers by hybrid electrospinning to prepare nanofibrous membranes with dual anti-adhesion properties.¹⁶³ The *in vivo* and *ex vivo* studies showed that the membranes loaded with ICA could successfully inhibit adhesion formation. In addition to the over-proliferation of fibroblasts, which can lead to scar and adhesion formation, inflammatory response and collagen synthesis are also key

factors in causing epidural scarring. Therefore, Shi and co-authors prepared a bilayer nanofibrous membrane loaded with different drugs in each layer by sequential electrospinning to reduce inflammation and prevent the extraordinary increase in collagen synthesis (Fig. 8a).¹⁶⁴ The main components of both layers are PCL and CS. The difference is that the mass ratio of PCL and CS is 8/2 and the loaded drug is MMC in the outer layer, while the inner PCL and CS have a mass ratio of 6/4 and the loaded drug is meloxicam (MX). The outer layer that contacts the fibrous tissue plays an important role in inhibiting the synthesis of DNA and collagen due to the effect of MMC. Meanwhile, benefiting from the anti-inflammatory property of MX, the inner layer contacting the spinal dura can prevent inflammation. The drug release profile showed that an initial burst release occurred in the first 5 days, followed by a steady and sustained release in the next 10 days, providing a promising anti-adhesion effect (Fig. 8b). The Rydell scores, representing the degree of epidural adhesion, showed that the MMC/MX dual-loaded group had the lowest adhesion scores, demonstrating that the combined anti-adhesion effect of MMC/MX was greatly improved (Fig. 8c).

Although this drug-loaded nanofibrous membrane has been well protected against epidural adhesions, this material cannot mimic the structural properties of graded tissues in the

human body. Developing bioscaffolds with hierarchical micro/nanofibrous structures to match the desired tissues holds great promise for biomimetic reconstruction and inducing the growth of matched cells and tissues. Therefore, Xu and co-authors prepared a biomimetic designed micro/nanoscale scaffold with integrated hierarchical dual fibrillar components to repair dura mater and prevent the formation of epidural scars *via* collagen molecule self-assembly, electrospinning, and biological interface crosslinking strategies (Fig. 8d and e).¹⁶⁵ In this biomimetic scaffolds, nano-scaled type I collagen (COL-I) fibers containing interconnected pores formed a network structure between electrospun silk fibroin (SF) or PLLA microfibers, which well mimics the natural extracellular matrix (Fig. 8f). The immunohistochemical analysis of α -SMA in the debrided epidural scar tissue showed that compared with the control group, α -SMA expression was reduced in the SF-COL group, which is a marker for the formation of myofibroblasts. It further indicated that the bionic scaffold could reduce the formation of epidural scars by regulating cell proliferation and differentiation to achieve excellent anti-adhesion effects (Fig. 8g).

5.2. Nanofibrous membranes for prevention of pericardial adhesion

Pericardial adhesions often occur after cardiac surgery and can significantly increase the rate of re-morbidity and mortality, which is a great challenge for patients with congenital heart disease.^{166,167} Therefore, the prevention of pericardial adhesions can largely reduce the number of surgeries and decrease the mortality rate of surgeries.¹⁶⁸ Currently, numerous materials have been developed for the prevention of pericardial adhesions, such as expanded polytetrafluoroethylene (e-PTFE) and bioabsorbable membranes (Serafilm).^{166,169} However, the e-PTFE membranes tend to cause severe inflammation due to their non-biodegradability and the bioabsorbable membranes degrade too quickly *in vivo*.^{170,171} Thus, the construction of new types of pericardial anti-adhesion materials is urgent.

Feng and co-authors prepared a GT/PCL hybrid nanofibrous membrane *via* co-hybrid electrospinning, which combines the good biological properties of GT with the slower degradation rate of PCL (Fig. 9a).¹⁷² This team's previous study showed that a good pericardial anti-adhesion effect can be obtained when the mass ratio of GT/PCL was 50/50. At 6 months postoperatively, it was evident from the dissected animals that little adhesion occurred in the heart, and the GT/PCL membrane fused with the pericardium to form a gross anatomical structure similar to the natural tissue. The thickness of the newly formed barrier was about four-times larger than that of the normal pericardium, limiting the penetration of fibroblasts into the surrounding tissues, and thus significantly reducing pericardial adhesions (Fig. 9b). Moreover, the histologic analysis further confirmed that with the gradual degradation of the GT/PCL membrane *in vivo*, a large number of host cells migrated into the material to form a natural barrier, which was structurally similar to the natural pericardium (Fig. 9c).

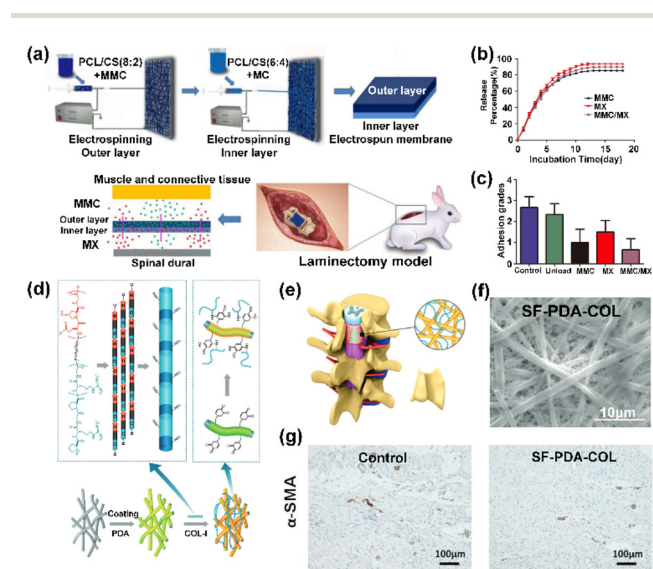


Fig. 8 Some representative electrospun nanofibers for epidural anti-adhesion. (a) Schematic illustration of preparation method and anti-adhesion application in laminectomy model of PCL-CS membrane. (b) Cumulative release of MMC/MX from PCL-CS membranes after incubation in PBS at 37 °C. (c) Gross evaluation of the adhesion formation in the control, non-loaded and MMC/MX-loaded membrane groups, $p < 0.05$ versus MMC/MX groups, and $p < 0.05$ versus control groups. Reprinted with permission from ref. 164. Copyright 2019, John Wiley and Sons. (d) Schematic illustration of the fabricated process of hierarchical micro/nanofibrous scaffolds. (e) *In vivo* spinal dural defect model. (f) Morphology of micro/nanofibrous scaffolds. (g) Comparison of expression of α -SMA in epidural scar tissues. Reprinted with permission from ref. 165. Copyright 2017, John Wiley and Sons.

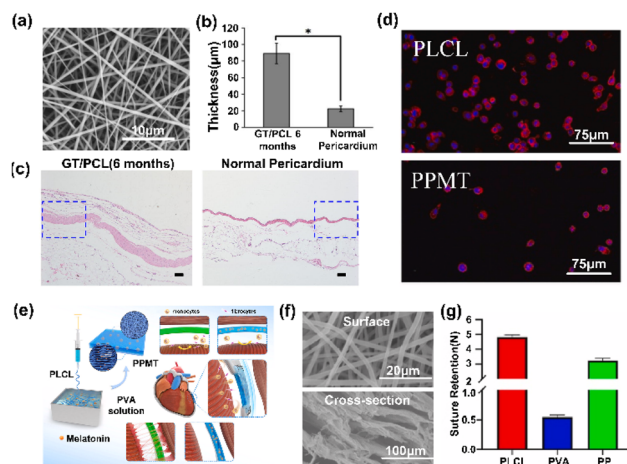


Fig. 9 Some representative electrospun nanofibers for pericardial anti-adhesion. (a) SEM of electrospun GT/PCL membrane. (b) Thicknesses of the GT/PCL membrane and normal pericardium. (c) H&E staining of the GT/PCL membrane and normal pericardium. (d) TRITC Phalloidin staining of macrophages on PLCL, PPMT after 48 h of differentiation. Reprinted with permission from ref. 172. Copyright 2021, KeAi. (e) Schematic illustration of the preparation process and application in cardiac surgery to prevent adhesion formation of the functional PVA aerogel-based membrane. (f) SEM images of surface and cross-section of PPMT. (g) Suture retention strength of different materials. Reprinted with permission from ref. 177. Copyright 2019, Elsevier.

However, despite the excellent anti-adhesion effect of this anti-pericardial adhesion membrane, there is still a long way to go if its clinical applications are to be realized.

Many materials have been used to prevent pericardial adhesion, such as drugs, hydrogels, and anti-adhesion membranes.^{169,171,173–176} However, few anti-adhesion materials are both functional and suturable for pericardial reconstruction. Therefore, Jin and co-authors developed a functional barrier membrane (named PPMT) with suturability *via* a modified electrospinning method, which consisted of poly(L-lactide-caprolactone) lactide (PLCL) nanofibers, poly(vinyl alcohol) (PVA) aerogel, and melatonin (Fig. 9e).¹⁷⁷ Melatonin can achieve anti-inflammatory effects by affecting the activity of inflammatory cells and reducing the release of pro-inflammatory factors.^{178,179} The PVA aerogel with macroporous structure is highly hydrophilic, which is a good drug-carrying system and has a certain anti-adhesive effect.^{180,181} The PLCL nanofibers enhanced the mechanical properties of the composites and ensured sutureability.^{182,183} PPMT has a special microstructure, as can be seen in Fig. 9f, which has randomly arranged nanofibers on its surface, while its cross-section exhibits a layered and macroporous aerogel structure. This structure ensures its mechanical properties and serves as an ideal drug-carrying platform for better anti-adhesion effects simultaneously. As the outer layer of the heart, the barrier membrane applied needs to have appropriate tensile strength, and also strong suture retention strength. By testing the suture strength of different groups (Fig. 9g), it was found that PLCL

has the highest suture strength, indicating that it has a suitable mechanical strength for use in suturing. In addition to this, the *in vitro* anti-inflammatory properties of the different membranes were evaluated, and fewer monocytes differentiated into macrophages on the PPMT membranes compared to the pure PLCL membranes, suggesting that the PPMT membranes have an excellent anti-inflammatory effect, which can reduce the formation of adhesions.

5.3. Nanofibrous membranes for prevention of meningeal adhesion

Desmoid decompression is a commonly used treatment in brain surgery. However, postoperative adhesions between the temporalis muscle and the dura mater usually occur, which can lead to brain damage, increased mortality, and other adverse outcomes. Therefore, it is important to use anti-adhesive membranes to minimize postoperative dural adhesions.^{184,185} Many anti-adhesive membranes have been used after tendon, spinal, abdominal, and cardiac surgeries, but information about the preventive of dural adhesions after craniectomy is relatively scarce.^{172,177,186–192} In addition to good barrier and excellent mechanical properties, anti-adhesive membranes used in the cranial cavity need to be stable in the human body for 3–4 months. Shi and co-authors prepared PCL/GT (PG) nanofibrous membranes with a prolonged degradation time by controlling the PCL/gelatin ratio *via* a hybrid electrospinning technique (Fig. 10a).¹⁹³ Previous studies showed that the membrane with a PCL/GT ratio of 50/50 (w/w) could only maintain its barrier function for 3–8 weeks. Thus, in this study, the ratio of PCL was increased to 60% (PG60-40) to prolong the degradation time of the membranes and maintain equilibrium. The obtained membranes showed a randomly interconnected structure with a porosity of 69–89%, which can facilitate nutrient-waste exchange (Fig. 10b). Moreover, due to the excellent mechanical properties of PCL, the prepared PG (60-40) membranes showed a stress strength of about 10 MPa, which is crucial for the effectiveness of anti-adhesive membranes that need to withstand the forces exerted during surgical procedures (Fig. 10c). Additionally, the nanofibrous membranes have unique advantages in preventing the invasion of fibroblasts as a physical barrier because of their small diameters (Fig. 10d). No cells reached the opposite side of the membrane, further indicating that the PG60-40 membrane has a good function of resisting fibroblast invasion (Fig. 10e). The degradation rate of the membrane also has a great influence on the anti-adhesion effect, and the rapid degradation rate of the membrane may lead to the loss of its protective effect during the critical period of adhesion occurrence, thus increasing the risk of adhesion. Compared with the representative commercial Bio-Gide® membrane that has a faster degradation rate (almost degraded at 12 weeks), the PG60-40 membrane still has a certain thickness at 24 weeks, which can act as a barrier for a longer period of time (Fig. 10f).

Another accompanying problem after debridement decompression is dural defect, which is a major problem in neuro-

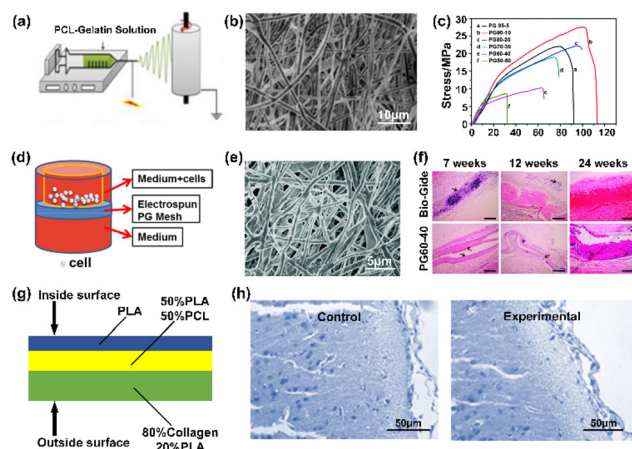


Fig. 10 Some representative electrospun nanofibers for meningeal anti-adhesion. (a) Schematic illustration of preparation method of electrospun PCL-gelatin hybrid membrane. (b) SEM of the obtained nanofibrous membranes with composition of PCL and gelatin: 60 : 40. (c) Stress-strain curve of electrospun PG membranes with different compositions of PCL and gelatin in the dry state. (d) Schematic diagram of barrier function characterization of PG membrane. (e) SEM micrograph of the opposite side of the cell-seeded PG 60-40 membrane for 3 days. (f) Histological micrographs with H&E staining (scale bar = 500 μm) of PG60-40 membranes after subcutaneous implantation. The arrows indicate the remaining membranes. Reprinted with permission from ref. 193. Copyright 2015, The Royal Society of Chemistry. (g) Structural diagram of the multiplayer dural substitute. (h) HE staining of brain tissue in the control group and in the experimental group after 30 days of implantation. Reprinted with permission from ref. 197. Copyright 2013, John Wiley and Sons.

surgery. Inadequate closure of the dura mater postoperatively may lead to complications, such as cerebrospinal fluid leakage and meningitis.^{194–196} Thus, there is a urgent need to develop new materials for replacing the dura mater and preventing adhesion. Wang and co-authors prepared a multi-layered scaffold that can reduce adhesion and promote dura repair using sequential electrospinning.¹⁹⁷ As shown in Fig. 10g, the inner layer of the scaffold facing the brain tissue was composed of PLA nanofibers to reduce tissue adhesion. The middle layer of the scaffold consisted of PCL/PLA composite nanofibers to provide a watertight seal to prevent cerebrospinal fluid leakage. The outer PLA/GT nanofiber layer containing a high amount of collagen can promote cell adhesion and proliferation. Results from *in vitro* studies and animal models showed that this multi-layered fibrous scaffold has sufficient mechanical strength and biochemical properties, which can enhance dural repair, while preventing dural adhesions. Moreover, the number of cells in the outer layer of the scaffold was significantly higher than that in the inner layer, indicating good biocompatibility and ability to promote cell growth on the outer surface (Fig. 10h). In summary, the preparation of dural substitutes with multiple functions and structures is promising in the field of neurosurgery, which may provide a reference for the preparation of other types of tissue regeneration biomaterials.

6. Conclusions and future perspectives

Tissue adhesions often lead to adverse consequences such as pain, dysfunction, and organ damage, which can significantly affect the quality of life and cause significant financial burden. Clinically, tissue adhesions manifest across a broad spectrum, involving structures such as tendons, dural sacs, the peritoneum and pericardium. Thus, to solve these problems, numerous materials have been explored to prevent the occurrence of tissue adhesions. Among these therapeutic materials, nanofibrous membranes have emerged as a highly promising anti-adhesion agent. These membranes not only serve as physical barriers to prevent adhesion formation but also act as drug carriers for controlled drug release. Moreover, these membranes can mimic human tissues for the better repair of damaged tissues. Furthermore, the desired properties of nanofibrous membranes can be achieved by altering their compositions, such as surface hydrophilicity, drug release mechanism, antibacterial capacity, degradation rates, and mechanical properties. The methodologies employed for the preparation and modification of nanofiber membranes are applicable to various other medical domains, including wound dressings, biosensors, and tissue engineering.^{198–200} These outstanding research works have facilitated the diversified development of nanofiber membranes in the medical field.

However, although much progress has been made in nanofibrous anti-adhesive membranes, several challenging issues still need to be addressed. Nevertheless, challenges coexist with opportunities, and recognizing the prospective development of the future can better assist in addressing current issues. Firstly, current drug-loaded nanofiber scaffolds lack sustained drug release for a sufficiently long period due to their high surface-to-volume ratios. They prevent adhesion during early healing but can lead to recurrence post complete drug release. Thus, to address this problem, more special structures can be created through the emulsion or coaxial electrospinning techniques. Additionally, drug-coated nanoparticles can be incorporated into nanofibers to extend the drug release duration. Secondly, the controlled release of drugs in drug-loaded anti-adhesive membranes to correspond with distinct tissue repair stages remains a challenge. An advanced smart drug delivery system that can sequentially release drugs with the effects of anti-inflammation, healing promoting and anti-adhesion can be further explored in the future. For instance, drug-loaded scaffolds can be designed with responsive capabilities to release distinct drugs in response to variations in environmental pH levels or growth factor concentrations. Thirdly, excessive drug concentrations on nanofibers can adversely affect normal tissue healing, such as impairing the intrinsic healing of tissues. Hence, precise regulation of the drug concentration is crucial. Additionally, an innovative avenue involves replacing pharmaceutical agents with the integration of nanofibrous membranes and stem cells, embracing a more biologically-driven approach to

address tissue adhesion. In this way, it not only promotes the *in situ* differentiation and growth of the desired cell types to facilitate the healing of specific tissues but also provides a favorable anti-adhesion effect, while mitigating the risks associated with drugs. Additionally, the most recent reports of complex nanostructures, and also electrospun/electrosprayed beads-on-a-string products are believed to be capable of providing useful solutions for this issue.^{201–208} Fourthly, anti-adhesive membranes must exhibit suitable mechanical properties. On one hand, it should facilitate facile manipulation and fixation within the human body during suturing. On the other hand, the anti-adhesion barrier must endure substantial tensile forces in regions such as tendons and the heart. Enhanced mechanical properties can significantly enhance the effectiveness of anti-adhesion measures. The integration of synthetic polymers with superior mechanical traits into the electrospinning process has emerged as a promising strategy to enhance the anti-adhesion efficiency. Furthermore, achieving both healing and anti-adhesion effects simultaneously is also a design trend in anti-adhesion materials.

With the continuous development of biomaterials and manufacturing technology, there will be more excellent materials and more advanced technology for preventing tissue adhesion. We believe that the era of “non-adhesive” will arrive soon with the increasing in anti-adhesion research.

Conflicts of interest

There are no conflicts to declare.

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